

Module - 5

Testing of Hypothesis

Branch of statistics which helps in decision making and it is initiated by J. Neyman and E. S. Pearson.

Statistical hypothesis

Assumptions or statements which may or may not be true about a population or equivalently about the probability distribution characterising the given population which we want to test on the basis of evidence from a random sample.

Simple hypothesis

If the hypothesis completely specifies the population, it is called simple hypothesis.

Composite hypothesis

If the hypothesis does not completely specify the population, it is called composite hypothesis.

Test of significance

A procedure to assess the significance of the statistics or difference between two independent statistics.

Null hypothesis

Hypothesis which is tested for possible rejection under the assumption that it is true. It is denoted by H_0 .

Alternate hypothesis

Any hypothesis which is complementary to null hypothesis. It is denoted by H_1 .

For example,

If $H_0 : \mu_1 = \mu_2$, then H_1 can be

$H_1 : \mu_1 \neq \mu_2$ (2 tailed alternative,
 $\mu_1 < \mu_2$ or $\mu_1 > \mu_2$)

$H_1 : \mu_1 < \mu_2$ (left tailed alternative)

$H_1 : \mu_1 > \mu_2$ (Right tailed alternative)

Types of errors in testing of hypothesis

In any test procedure, the four possible mutually disjoint and exhaustive decisions are

1. Reject H_0 when it is false.
2. Accept H_0 when it is true
3. Reject H_0 when it is true
4. Accept H_0 when it is false

From the above, 3 & 4 are errors.

Type 1 Error

Rejecting a true null hypothesis

Type 2 Error

Accepting a wrong null hypothesis

$$P(\text{Reject } H_0 \text{ when it is true}) = P(\text{Type 1 Error}) = \alpha$$

$$P(\text{Accept } H_0 \text{ when it is false}) = P(\text{Type 2 Error}) = \beta$$

Here α and β are the sizes of type 1 error and type 2 error respectively.

In terminology of industrial quality control while inspecting the quality of a manufactured lot,

$\alpha = P(\text{rejecting a good lot}) = \text{Producer's risk}$

$\beta = P(\text{accepting a bad lot}) = \text{Consumer's risk}$

It is not possible to minimise these errors simultaneously.

Level of Significance

The maximum size of type I error which we are prepared to risk is called level of significance. It is denoted as α .

Commonly used levels of significance are 5% and 1%.

i.e., when we test the hypothesis, we can never be 100% certain about our conclusion. We can only be confident to a certain level of fully, a high level. Typically we construct our test so that we will be 95% certain that the conclusion we draw is a correct one. This is called 95% confidence level or 5% significance level.

If we adopt 5% significance level, it implies that in 5 samples out of 100 which are likely to reject a current H_0 . Level of significance is fixed in advance before collecting the sample information.

Critical Region

Suppose we take several samples of same size from a given population and compute some statistic t for each of these samples.

Let $t_1, t_2, \dots, t_k$ be values of statistic for these samples, each of these values may be used to test some null hypothesis H_0 . Some values may lead to rejection of H_0 , while others may lead to acceptance of H_0 . These sample statistics $t_1, t_2, \dots, t_k$ comprising the sample size may be divided into two mutually disjoint jobs, one leading to the rejection of H_0 and other leading to the acceptance of H_0 .

The statistic which leads to rejection of H_0 gives us a region called critical region (C) or rejection region (r). While the need for

acceptance of H_0 gives us a region called acceptance region (A).

Thus if statistic 't' belongs to 'c', H_0 is rejected.

and if $t \in A$, H_0 is accepted.

One tail and Two tailed Test

A test of any statistical hypothesis where the alternative hypothesis is one tail is called a one-tail test.

eg:- If $H_0: \mu_1 = \mu_2$, then H_1 can be

$$H_1: \mu_1 < \mu_2 \text{ or}$$

$$H_1: \mu_1 > \mu_2$$

Here H_1 is a one tail (single tail) test.

A test of any statistical hypothesis where the alternative hypothesis is two tail such that $H_0: \mu_1 = \mu_2$,

$$\text{Then } H_1: \mu_1 \neq \mu_2$$

Here H_1 is an example for two tailed test.

i.e., either $\mu_1 < \mu_2$ or $\mu_1 > \mu_2$.

Procedure for testing of hypothesis

Step 1 : Setup the null hypothesis.

Step 2 : Setup the alternative hypothesis.

Step 3 : Choose the appropriate level of significance.

Step 4 : Define and compute the test statistic.

Step 5 : Obtain the critical value and critical region of the test statistic from appropriate tables.

Step 6 : If the computed value of test statistic lies in the rejection region, we reject H_0 at level of significance.

Step 7 : Write the conclusion.

Note

- If the calculated value of test statistic is greater than table value, then we can say that it is significant and the null hypothesis H_0 is rejected.

- If the calculated value of test statistic is less than table value, then we can say that it is not significant and the null hypothesis H_0 is accepted.

- Samples drawn may be large or small. In practical point of view, a sample is termed as small if $n \leq 30$.

Various sample tests are,

Small Sample Test

1. Student's 't' test for single mean.
2. Student's 't' test for difference of means.
3. Paired 't' Test.
4. χ^2 Test
5. F Test

Large Sample Test

1. Test of significance for a single mean.
2. Test of significance for difference of means.

SMALL SAMPLE TEST

1. Test for single Mean

- If we are interested to test for single mean;
- If the given normal population has a specified value of population mean.
 - The sample mean $\bar{x}$ differs significantly from the specified value of population mean.
 - If a given random sample $x_1, x_2, \dots, x_n$ of size 'n' has been drawn from a normal population with specified mean.

Steps

1. Setup null hypothesis, H_0 .
2. Setup alternative hypothesis, H_1 .
3. Choose level of significance.
4. Test statistic $t = \frac{\bar{x} - \mu}{\sqrt{s^2/n}}$, where $\bar{x}$ sample mean, μ population mean.

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

The test statistic follows 't' distribution with $(n-1)$ degrees of freedom.

5. The test statistic, ' t ' is compared with table value of ' t ' with $(n-1)$ degree of freedom at the given level of significance.
6. If the calculated value of ' t ' is greater than table value of ' t ', we can say that test is significant and H_0 is rejected.
If the calculated value is less than the table value, H_0 is accepted.
7. Based on this, find the conclusion.